Hyper-Realistic Digital Twin and Software-in-the-Loop Architectures for Real-Time AC Microgrid Simulation on Embedded RTOS

Introduction to Embedded Electromagnetic Transient Simulations

The evolution of modern power systems necessitates the deployment of highly advanced testing methodologies to ensure the stability and reliability of grid-edge devices, protective relays, and distributed energy resources. Controller Hardware-in-the-Loop (CHIL) and Software-in-the-Loop (SIL) environments have emerged as critical paradigms for validating these systems. Historically, the simulation of electromagnetic transients required the computational prowess of mainframe computers or specialized, high-cost Field Programmable Gate Arrays (FPGAs) running advanced Electromagnetic Transients Programs (EMTP). Foundational literature in power system stability and transient analysis—such as the authoritative texts by Arrillaga on power system harmonics, Kundur on power system stability, and Machowski on system dynamics—established the continuous-time physics that govern these networks. However, translating these complex, differential phenomena into a hyper-realistic 'Digital Twin' capable of executing within the strict deterministic deadlines of a Real-Time Operating System (RTOS) on a commercial, resource-constrained microcontroller, such as the ESP32, demands a profound paradigm shift in numerical methods.

This comprehensive report details the extraction, synthesis, and algorithmic optimization of discrete-time mathematical models necessary for real-time AC microgrid simulation, specifically targeting a 230V, 50Hz topology. The analysis systematically deconstructs the mathematical transition from continuous-time physics to discrete-time state-space equations using trapezoidal integration and Modified Nodal Analysis (MNA). Furthermore, it provides the exact mathematical formulations required to generate synthetic Analog-to-Digital Converter (ADC) readings encompassing steady-state fundamental waveforms, IEEE 519-2022 compliant harmonic distortion, and additive Gaussian white noise.

Beyond steady-state synthesis, this report exhaustively models transient network phenomena. It captures the highly non-linear physics of induction motor inrush currents, specifically detailing the exponential decay of the DC offset superimposed upon a slip-dependent AC envelope. To test protective relays effectively, the report formulates lightweight, computationally bounded mathematical models of Current Transformer (CT) core saturation, demonstrating how secondary waveforms clip during severe short-circuit faults. The dynamic introduction of power quality anomalies, including voltage sags, swells, and low-frequency flicker, is also rigorously defined. Finally, the report explores architectural optimizations critical for C++ embedded execution, contrasting Float32 hardware Floating-Point Unit (FPU) utilization against Float64 software emulation, and analyzing the algorithmic superiority of Direct Digital Synthesis (DDS) and recursive coupled-form oscillators over standard trigonometric function calls.

Theoretical Foundations: EMTP and Discrete-Time State-Space Formulation

The accurate simulation of a power grid requires the continuous evaluation of energy storage elements, specifically inductors and capacitors, which are mathematically governed by ordinary differential equations. In an embedded execution environment, these continuous-time equations cannot be solved analytically in real-time. Instead, numerical integration techniques must be applied to step through time at a fixed discrete interval, denoted as Δt.

The Dommel Companion Circuit Methodology

The foundational mathematics of modern transient simulators rely on the methodology pioneered by Hermann Dommel in the 1960s at the Technical University of Munich and later at the Bonneville Power Administration (BPA). Dommel's approach utilizes the trapezoidal integration rule to convert continuous-time lumped circuit elements into discrete-time "companion circuits". The trapezoidal rule is favored in EMTP applications because of its absolute numerical stability and its ability to avoid the spurious numerical oscillations that often plague forward Euler or higher-order Runge-Kutta methods when dealing with stiff power system equations.

For a fundamental lumped inductor, the relationship between voltage and current is defined continuously as $v_L(t) = L \frac{di_L(t)}{dt}$. When discretized using the trapezoidal integration method over a time step Δt, the equation is algebraically rearranged into a Norton equivalent companion circuit. This companion circuit consists of a constant equivalent conductance (g_L) placed in parallel with a known history current source ($I_{hist, L}$). The discrete-time update equation for the inductor current at the current time step is formulated as follows:

The equivalent conductance and the history current term, which encapsulates the state of the inductor from the previous time step, are defined explicitly as:

Similarly, for a lumped capacitor, the continuous-time differential equation $i_C(t) = C \frac{dv_C(t)}{dt}$ is subjected to the same trapezoidal discretization. The resulting Norton equivalent companion circuit equation is:

The equivalent conductance and the history current term for the capacitor are defined as:

By transforming every energy storage element into a parallel combination of a simple resistor and a known current source, the entire differential complexity of the microgrid is reduced to a purely resistive DC network at each individual time step.

Descriptor State Space Equations and Modified Nodal Analysis (MNA)

While standard state-space models in the form of $\dot{x} = A \cdot x + Bu$ are conceptually elegant, forming them for large, dynamically switching power networks requires complex topological tree and co-tree sorts that are computationally prohibitive for an ESP32 microcontroller. Instead, real-time EMTP solvers construct Descriptor State Space Equations (DSE) utilizing Modified Nodal Analysis (MNA).

Theoretical investigations prove that the discrete descriptor state space equations derived via MNA are mathematically identical to the assembled companion circuit models in classical

EMTP. The MNA approach partitions the network into element matrices based on a node incidence matrix, yielding a global system update equation that is solved at every RTOS tick:
$$ \cdot [e(t)] = + [I_s(t)]$$
In this matrix formulation, $$ represents the global nodal admittance matrix, $[e(t)]$ is the vector of unknown node voltages to be calculated, $[I_{his}]$ contains the aggregated history currents from all storage elements, and $[I_s(t)]$ represents the active, independent current sources injecting into the grid.
Because the time step Δt is strictly fixed in real-time embedded simulations to maintain deterministic execution, the admittance matrix $$ remains completely constant as long as the network topology (e.g., switch states or breaker positions) does not change. This provides a massive computational advantage for the ESP32. During the RTOS initialization phase, the inverse of the admittance matrix, $^{-1}$, can be pre-computed and stored in memory. Consequently, the real-time processing loop is reduced from an $O(N^3)$ Gaussian elimination problem to a highly efficient $O(N^2)$ simple matrix-vector multiplication, ensuring the simulation effortlessly meets microsecond-level deadlines.

Steady-State Synthesis: Fundamental Waveforms and Harmonic Distortion

Creating a hyper-realistic Digital Twin of a 230V, 50Hz microgrid requires acknowledging that ideal, pure sinusoidal waveforms do not exist in physical power systems. The proliferation of power electronics, variable frequency drives, and solid-state lighting introduces continuous harmonic pollution into the grid. A robust SIL simulation must synthesize these non-linear artifacts accurately to test the true response of measurement algorithms and protective relays.

Compliance with IEEE 519-2022 Harmonic Standards

The generation of synthetic steady-state signals must strictly conform to the regulatory limits defined by the IEEE 519-2022 Standard for Harmonic Control in Electric Power Systems. This standard specifies rigorous acceptable limits for voltage and current distortion at the Point of Common Coupling (PCC), which serves as the electrical boundary between the utility operator and the consumer.
For a 230V AC microgrid, the system falls under the low-voltage classification ($V \le 1.0 \text{ kV}$). According to the standard, the Total Harmonic Distortion (THD) for voltage in this tier must not exceed 8.0%, and no individual harmonic voltage shall exceed 5.0% of the fundamental frequency.

Bus Voltage at PCC	Individual Harmonic Voltage Limit (%)	Total Harmonic Distortion (THD) Limit (%)
$V \le 1.0 \text{ kV}$	5.0	8.0
$1 \text{ kV} < V \le 69 \text{ kV}$	3.0	5.0
$69 \text{ kV} < V \le 161 \text{ kV}$	1.5	2.5
$V > 161 \text{ kV}$	1.0	1.5

Furthermore, the IEEE 519-2022 standard defines Current Distortion Limits based on the short-circuit strength of the network, mathematically expressed as the ratio of maximum short-circuit current to maximum demand load current (I_{SC}/I_L). A "stiff" microgrid with a high ratio is permitted to absorb more harmonic current without degrading the global voltage profile, whereas a "weak" microgrid (ratio < 20) is held to stringent Total Demand Distortion (TDD) limits.

Short Circuit Ratio (I_{SC}/I_L)	Limit for Harmonics $2 \le h < 11$ (%)	Limit for Harmonics $11 \le h < 17$ (%)	Total Demand Distortion (TDD) Limit (%)
< 20	4.0	2.0	5.0
20 < 50	7.0	3.5	8.0
50 < 100	10.0	4.5	12.0
100 < 1000	12.0	5.5	15.0

To simulate these limits dynamically, the microcontroller synthesizes the instantaneous voltage $v(t)$ as a discrete-time Fourier series. The waveform superimposes the fundamental 50Hz sine wave with the 3rd, 5th, and 7th harmonics. These specific odd harmonics are intentionally selected because they dominate ungrounded or unbalanced three-phase distribution networks; the 3rd harmonic generates severe zero-sequence currents that overheat neutral conductors, while the 5th harmonic creates negative-sequence counter-torque in rotating machines.

The mathematical synthesis equation processed at each Δt is:

Where f_0 is the fundamental 50Hz frequency, V_{rms} is the nominal 230V rating, and the scaling coefficients α_h are constrained strictly below 0.05 (representing 5.0%) to ensure rigorous compliance with the IEEE 519-2022 voltage limits. The term $\eta(t)$ represents the continuous injection of additive background noise.

Stochastic Sensor Noise: Additive Gaussian White Noise (AWGN)

Real-world ADC hardware is inherently imperfect. Measurements derived from physical current and voltage sensors contain thermal noise, electromagnetic interference, and digital quantization errors. A hyper-realistic SIL simulation must replicate these artifacts; otherwise, the tested algorithms may appear artificially robust.

Simply invoking the standard C++ rand() function is mathematically insufficient for modeling sensor noise. The rand() function produces a uniform probability distribution, which does not accurately represent the physical mechanics of thermal and electrical noise. Physical noise follows a normal, or Gaussian, distribution characterized by a distinct bell curve. To map uniformly distributed pseudo-random integers into an accurate Gaussian distribution, the discrete-time model must employ the Box-Muller transform.

The Box-Muller Transform Algorithm

The Box-Muller method takes two independent, uniformly distributed continuous random variables on the interval (0, 1], denoted as U_1 and U_2, and mathematically maps them into two independent, standard, normally distributed random variables, z_0 and z_1, which exhibit a mean of zero and a standard deviation of one.

The exact mathematical formulation of the transform is:

Once the standard normal variables are generated, the final synthetic noise sample $\eta(t)$ injected into the grid simulation is scaled by the desired noise standard deviation (σ), which defines the width of the noise floor, and shifted by a mean (μ), which can simulate DC offset bias in the analog sensors.

Embedded Implementation Strategies

Executing the Box-Muller transform is computationally demanding for an embedded processor because it relies on logarithm, square root, and trigonometric function evaluations. To optimize this for an ESP32 running an RTOS, two primary strategies must be deployed.

First, the underlying uniform random numbers should not be generated by the slow and statistically weak rand() function. Instead, a Tausworthe Uniform Random Number Generator (URNG) or a fast Linear Feedback Shift Register (LFSR) should be utilized, as these provide superior statistical randomness and execute in a fraction of the clock cycles.

Second, the Box-Muller transform inherently produces two valid Gaussian samples (z_0 and z_1) per iteration. To halve the average computational burden, the C++ implementation must cache the value of z_1 in a static or class-member variable during the first execution. On the subsequent RTOS tick, the algorithm simply retrieves the cached z_1 value instead of recalculating the entire transcendental equation block.

Transient Inrush Physics: Induction Motor Dynamics

Induction machines constitute the vast majority of industrial loads within modern AC microgrids. The dynamic behavior of an induction motor during startup is a critical transient event that severely impacts the entire electrical network. When an induction motor undergoes a Direct-On-Line (DOL) start, it behaves momentarily like a transformer with a short-circuited secondary winding, drawing a massive inrush current that typically peaks at 6 to 10 times its nominal steady-state operating current.

This transient inrush phenomenon is not a simple amplified sine wave; it is a highly asymmetrical waveform characterized by complex electro-mechanical interactions between the stator and the rotor. The transient starting current fundamentally consists of an exponentially decaying DC offset superimposed onto a slip-dependent AC current envelope.

The Exponential DC Offset Decay

At the exact instant the motor breaker closes, the magnetic flux linkages within the inductive stator and rotor windings cannot change instantaneously. To satisfy the fundamental boundary conditions of the governing differential equations, a unidirectional DC current component is instantly generated.

The magnitude of this initial DC offset is entirely dependent on the specific electrical angle of the 50Hz supply voltage at the millisecond the switch closes. If the breaker closes when the voltage is at a zero-crossing, the DC offset is maximized, resulting in severe waveform asymmetry. This DC component decays exponentially over time, governed by the inherent L/R time constant (τ) of the motor's stator network:

The presence of this DC offset is responsible for the extreme mechanical torque pulsations experienced during the first few cycles of acceleration, which can exceed the nominal operational torque by a factor of four to five. Furthermore, if the stator power supply is suddenly disconnected during operation, the collapsing magnetic field induces a stator back-EMF. This decay process can be mathematically captured by the expression $e_s = k \left(\frac{1}{T_{2r}} + \omega_r \right) e^{-2t/T_r}$, which allows sensorless speed identification algorithms to calculate the rotor time constant T_r.

The Slip-Dependent AC Envelope and Current Displacement

The AC component of the motor's starting current does not remain static; it forms an envelope that slowly decays as the rotor accelerates from a complete standstill to its final synchronous speed. This dynamic acceleration alters the relative frequency of the magnetic field cutting across the rotor bars, a parameter formally defined as the motor slip (s).

A conventional steady-state equivalent circuit model fails to capture the true transient profile because it assumes constant resistance and inductance values. A hyper-realistic Digital Twin must account for the "skin effect" or "current displacement" within the rotor slots. At the moment of a cold start ($s = 1$), the frequency of the induced currents within the rotor bars equals the full 50Hz grid frequency. This high frequency forces the alternating current to migrate toward the outer periphery of the rotor bars, significantly increasing the effective active resistance and simultaneously decreasing the leakage reactance.

As the motor accelerates and approaches synchronous speed ($s \to 0$), the frequency of the rotor currents drops to near zero (slip frequency), allowing the current to distribute uniformly across the entire cross-section of the bar. The analytical mathematical formulas required to dynamically calculate the slip-dependent rotor active resistance (R'_{2s}) and rotor reactive resistance (X'_{2s}) are provided as follows :

Where:

- R'_{2N} and X'_{2N} are the nominal steady-state parameters under normal load.
- $R'_{2,1}$ is the dramatically higher rotor resistance at standstill ($s=1$).
- X'_{2a} represents a baseline inductive reactance, and b is a geometrical design constant of the specific rotor cage.

By continuously recalculating these parameters based on the integrated mechanical speed of the rotor at every Δt, and updating the global admittance matrix $$, the simulation accurately reproduces the rapidly decaying AC current envelope. Models that ignore current displacement consistently underestimate the starting torque and overestimate the duration of the inrush current, leading to faulty relay coordination settings.

Sensor Nonlinearities: Lightweight Current Transformer (CT) Saturation Models

In the domain of SIL protective relay testing, the accurate simulation of primary power flows is only half the challenge; the Digital Twin must also simulate the physical limitations of the measurement sensors. Current Transformers (CTs) utilize ferromagnetic steel cores to step down massive grid currents to safe measurement levels (typically 1A or 5A secondary). However, when subjected to severe short-circuit faults containing high DC offsets, the magnetic flux inside the CT core rapidly accumulates in a unidirectional manner.

When the core reaches its maximum flux density, all magnetic dipoles become aligned. At this point, the CT enters "saturation." During saturation, the core can no longer induce a secondary voltage, causing the secondary current waveform to severely clip, distort, and drop toward zero. This phenomenon produces harmonic-rich, sawtooth-like waveforms that frequently cause differential relays to misoperate, making it a mandatory feature for realistic testing.

While advanced metallurgical models like the Jiles-Atherton theory of ferromagnetic hysteresis provide extreme accuracy, their complex differential equations require iterative, variable-step solvers that are catastrophic for fixed-step RTOS execution. A hyper-realistic embedded simulator must employ mathematically lightweight alternatives.

The Volt-Time Area Concept

A highly efficient computational approach treats the CT core as a threshold-based "volt-time switch." This model calculates the accumulation of magnetic flux continuously by integrating the excitation voltage (V_x) over time. The excitation voltage is defined as $V_x = I_S \times (R_{CT} + Z_B)$, where I_S is the secondary current, R_{CT} is the internal winding resistance, and Z_B is the connected external burden.
The mathematical flux accumulation is directly proportional to the integrated area under the voltage curve:
As long as this accumulated flux remains below the physical saturation limit of the core (ϕ_S), the CT operates in its linear region, and the secondary current is a perfect scaled replica of the primary current. However, if the integral exceeds the saturation threshold, the "switch" activates, shunting the current into the magnetizing branch. The instantaneous secondary voltage during this saturated state collapses and can be geometrically calculated as :
where $V_S = 2\phi_S N$ (with N representing the number of turns). This method avoids differential equations entirely, using discrete geometric boundary limits to clip the simulated ADC output.

The Fröhlich Equation and Arctangent Approximations

For simulations requiring a smooth, continuous transition through the "knee point" of the B-H curve rather than a binary switch, the Fröhlich equation is historically utilized to emulate the S-shaped saturation characteristic. It calculates the instantaneous magnetizing inductance (L_M) based on the incremental permeability (μ_r) of the steel :
However, updating L_M dynamically means the Y_{emt} matrix must be inverted at every single time step, which stalls the ESP32 CPU. To preserve the computational advantage of a pre-inverted, constant admittance matrix, the non-linear magnetizing branch is removed from the matrix and modeled externally as a dependent current source injected into the $[I_s(t)]$ vector.
To calculate this non-linear current without complex iterations, an Arctangent curve-fitting function is employed to model the B-H relationship. The magnetizing current i_m as a function of instantaneous flux λ can be smoothly described using just three calibrated parameters :
Alternatively, a fractional inverse tangent model can be used to fit the exact knee-point geometry :
Because the ESP32 Floating-Point Unit (FPU) can evaluate arctangent functions highly efficiently, this mathematical approach provides an exceptional balance between physical accuracy and deterministic RTOS execution speed.

Power Quality Anomalies: Sags, Swells, and Flicker

Grid anomalies disrupt the stability of sensitive load equipment and trigger protective mechanisms. A complete microgrid Digital Twin must dynamically introduce these disturbances via mathematical modulation of the fundamental steady-state matrix injections.

Voltage Sags and Swells Formulation

A voltage sag (or dip) is a temporary reduction in the Root Mean Square (RMS) voltage magnitude, dropping to between 0.1 and 0.9 per unit (p.u.) for a duration ranging from a half-cycle to one minute. Sags are predominantly caused by massive inrush currents from remote motor starts or severe short-circuit faults pulling down the grid voltage. Conversely, a voltage swell is a sudden increase in RMS voltage up to 1.1–1.8 p.u., typically resulting from the sudden disconnection of a large load (load rejection) or from phase-to-ground faults causing the un-faulted phases to experience temporary overvoltages.
To simulate these discrete-time events seamlessly without disrupting the continuity of the phase angle, mathematical formulations employ Heaviside unit step functions $u(t)$ to trigger the exact onset time (t_1) and clearance time (t_2). The consolidated equation for an instantaneous sag or swell is:
Where A is the nominal peak amplitude (e.g., $230\sqrt{2}$ V), and α dictates the severity depth of the sag or the height of the swell ($0.1 \le \alpha \le 0.9$). A negative operator ($-\alpha$) generates a sag, while a positive operator ($+\alpha$) generates a swell.

Voltage Flicker Formulation

Voltage flicker represents a continuous, low-frequency fluctuation in the amplitude of the grid voltage. It is generated by highly cyclical, non-stationary loads drawing rapidly varying reactive power, such as arc furnaces, heavy welding equipment, or large mechanical shredders. The physical consequence of this fluctuation is the visible flickering of light sources, which causes significant physiological discomfort and psychological tiredness in humans.
Mathematically, flicker is pure amplitude modulation (AM) superimposed onto the 50Hz carrier wave. The discrete-time synthesis equation is defined as :
Where A_0 is the fundamental amplitude, f_0 is the 50Hz grid frequency, and m is the flicker modulation index (typically constrained between 0.005 and 0.02). The variable f_M represents the low-frequency modulation rate. According to the International Electrotechnical Commission (IEC) 61000-4-15 standard, which dictates flickermeter architecture, the human ocular and neurological system is most acutely sensitive to fluctuations occurring at exactly **8.8 Hz**. Therefore, locking $f_M = 8.8 \text{ Hz}$ provides the most stringent and realistic stress test for algorithms designed to compensate for power quality degradation.

Embedded C++ Optimization Strategies for ESP32 RTOS

Deploying complex matrix mathematics and thousands of sinusoidal evaluations per second on an ESP32 microcontroller requires extreme optimization. The ESP32 utilizes the Tensilica Xtensa architecture, which features specific hardware constraints and advantages that must dictate the C++ software architecture.

Float32 vs. Float64 Execution and FPU Utilization

A critical design choice in embedded EMTP development is the selection of numerical precision. In PC-based software like MATLAB or PSCAD, equations default to 64-bit double precision (Float64 or double). However, the ESP32 is equipped with a hardware Floating-Point Unit (FPU)

that natively supports only 32-bit single-precision mathematics (Float32 or float).
If a C++ algorithm utilizes a double variable or relies on standard trigonometric functions without explicit float suffixes, the ESP32 compiler will bypass the hardware FPU entirely. Instead, it will utilize software emulation libraries provided by the C standard library to perform the 64-bit calculations. Software emulation requires dozens, sometimes hundreds, of CPU clock cycles per arithmetic operation, causing catastrophic latency that will instantly violate the RTOS microsecond deadlines. Conversely, operations compiled for the Float32 FPU execute natively in just 1 to 2 clock cycles.
To ensure the hardware FPU is utilized, all mathematical library calls must explicitly invoke single-precision functions (e.g., sinf(), cosf(), sqrtf()), and all numerical constants must be suffixed with an 'f' (e.g., 3.14159f). While Float32 reduces the dynamic range and introduces minor truncation errors over millions of cycles compared to Float64, careful scaling of the MNA admittance matrix allows the simulation to remain perfectly stable.

Direct Digital Synthesis (DDS) and Phase Accumulators

Calculating the Taylor series approximations required by the sinf() function for the fundamental 50Hz wave, the 3rd, 5th, and 7th harmonics, and the low-frequency flicker modulation at every time step is highly inefficient. A fundamentally faster methodology is Direct Digital Synthesis (DDS).
A DDS architecture replaces trigonometric math with integer addition and memory lookups. The core of the DDS is the Phase Accumulator, an N-bit unsigned integer register that rolls over automatically, natively simulating the 2π periodic wrap-around of a circle. At every sample clock interval (f_c), a Tuning Word (M) is added to the accumulator. The frequency of the output waveform is precisely controlled by the tuning equation:
Instead of evaluating a sine function, the most significant bits (MSBs) of the phase accumulator are truncated and used as a direct index into a pre-computed Sine Lookup Table (LUT). For example, a 32-bit accumulator might use only the top 12 bits to index a 4096-element array. This truncation adds a minimal, acceptable level of phase noise but allows the processor to generate complex waveforms using only one integer addition and one memory read per harmonic. However, developers must ensure the LUT is small enough to fit within the ESP32's fast data cache, as cache misses pulling data from external PSRAM will nullify the speed advantage of DDS.

Recursive Oscillators: Coupled-Form and Goertzel Algorithms

To eliminate memory cache misses entirely while still bypassing the slow sinf() function, recursive mathematical oscillators generate sine waves purely through iterative multiplication.
The **Coupled-Form Oscillator** (often called the magic circle or quadrature oscillator) computes the next instantaneous waveform sample by rotating the previous sample's coordinates through a 2-D transformation matrix. To generate a continuous frequency with an angular phase step of $\theta = 2\pi f_{grid} \Delta t$, the discrete-time state-space equations are :
In C++, $\cos(\theta)$ and $\sin(\theta)$ are treated as immutable constants calculated only once during the device's boot sequence. Consequently, the real-time generation of a sine wave requires only two extremely fast FPU multiplications and one addition per step. To counteract long-term amplitude drift caused by finite Float32 rounding errors, the algorithm periodically applies a lightweight normalization factor to enforce the identity $x_1^2 + x_2^2 = 1$.
Finally, for SIL scenarios requiring the algorithmic extraction and detection of specific harmonic

amplitudes (e.g., simulating a protective relay filtering a fault), the **Goertzel Algorithm** provides an immensely optimized alternative to the Fast Fourier Transform (FFT). The Goertzel algorithm operates as a highly tuned Infinite Impulse Response (IIR) filter that computes individual terms of the Discrete Fourier Transform (DFT) recursively. By isolating specific frequencies of interest (e.g., exactly 50Hz, 150Hz, and 250Hz) rather than processing the entire wideband spectrum, the Goertzel filter drastically minimizes CPU cycle consumption, enabling real-time frequency assessment on the ESP32.

Conclusion

The pursuit of developing hyper-realistic AC microgrid Digital Twins on commercial RTOS microcontrollers marks a significant leap in accessible power system testing. By deliberately transforming complex continuous-time physics into discrete-time matrices utilizing Hermann Dommel's trapezoidal companion circuits and Modified Nodal Analysis, the computational burden is fundamentally shifted from iterative differential solvers to highly deterministic, real-time algebraic multiplications. The integration of advanced phenomenological mathematics—such as slip-dependent exponential equations for induction motor inrush, arctangent-based approximations for non-linear CT saturation, and rigorous IEEE 519-2022 compliant harmonic synthesis algorithms—ensures that the simulated electrical environment faithfully replicates the chaotic severity of physical power grids. Ultimately, capitalizing on processor-specific C++ optimizations, including enforcing native Float32 FPU execution, deploying the Box-Muller transform for AWGN, and replacing heavy trigonometric functions with Direct Digital Synthesis and recursive coupled-form oscillators, guarantees that these sophisticated analytical models execute seamlessly within strict microsecond deadlines.

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